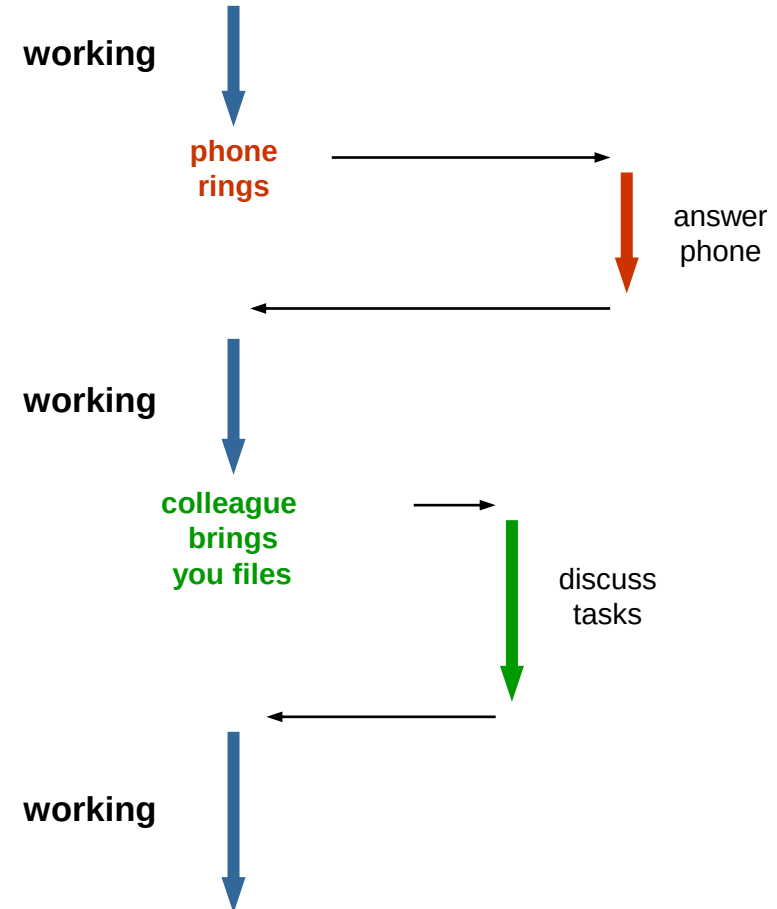
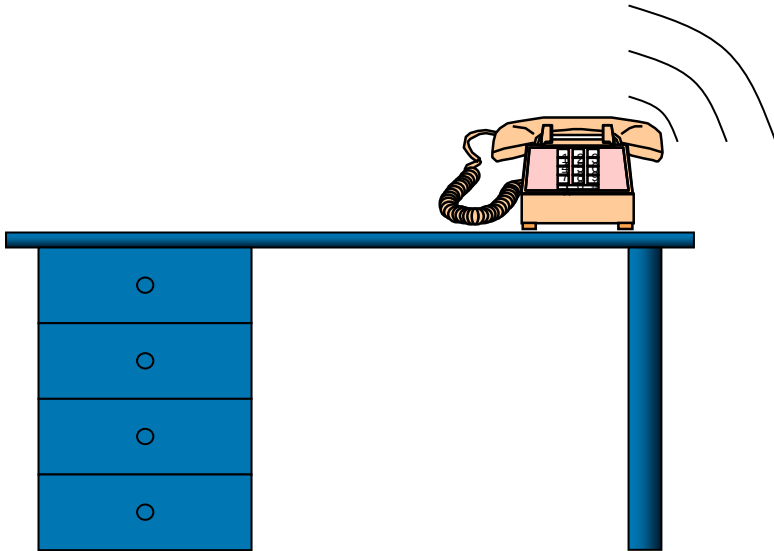


Exceptional Control Flow

Computer Engineering 1

Motivation

- You are working at your desk!



Motivation

- **Computers work concurrently on several tasks**
 - Often wait until result is available, e.g. network request
 - Fill idle time with useful tasks → efficient use of processing power
 - Interrupts signal, that result of a request is available
 - Requires switching between programs
- **Unexpected events – exceptions**
 - Division by 0, unaligned accesses, etc.
 - Must be caught and corresponding actions need to be triggered
- **Time critical events with high priority**
 - Interrupt running program → process event

Agenda

- **Polling**
- **Interrupt-Driven I/O**
- **Exceptions Cortex-M3/M4**
- **Interrupt-System Cortex-M3/M4**
- **Vector Table**
- **Storing the Context**
- **External Interrupt Pins**
- **Interrupt Control**
- **Nested Exceptions**
- **Special Interrupt Situations**
- **CMSIS Functions**
- **Data Consistency**

The lecture does not cover all the features of Cortex-M3/4 exception processing, in particular the following simplifications apply:

- No distinction between handler mode and thread mode
- No distinction between preempt priority and sub-priority fields
- Tail chaining and late arrival are not covered
- Instructions that are abandoned and restarted/resumed after interrupt service are not covered (e.g. LDM)

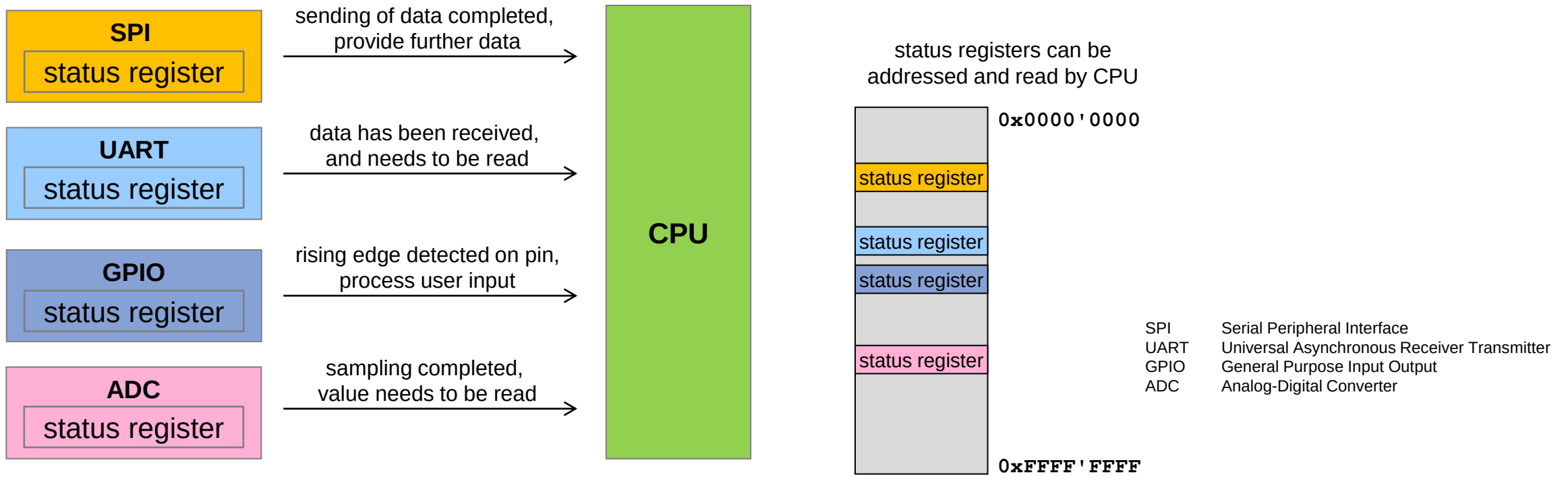
Learning Objectives

At the end of this lesson you will be able

- to explain advantages and disadvantages of polling and interrupt-driven I/O
- to distinguish the different types of exceptions on a Cortex-M3/M4
- to explain how the Cortex-M3/M4 recognizes and processes exceptions
- to explain the vector table of the Cortex-M3/M4
- to understand the basic functionality of the Nested Vectored Interrupt Controller (NVIC)
 - to enable and disable interrupts
 - to set and clear interrupts by software
 - to prioritize exceptions
 - to know how programmed priorities influence preemption of service routines
 - to explain how simultaneously pending interrupts are processed
- to implement a simple interrupt service routine in Cortex-M assembly
- to explain potential data consistency issues due to interrupts and to give potential examples

CPU Has to Act on Events

- **Peripherals signal events to CPU**
 - Something happened that requires a CPU service → call a subroutine

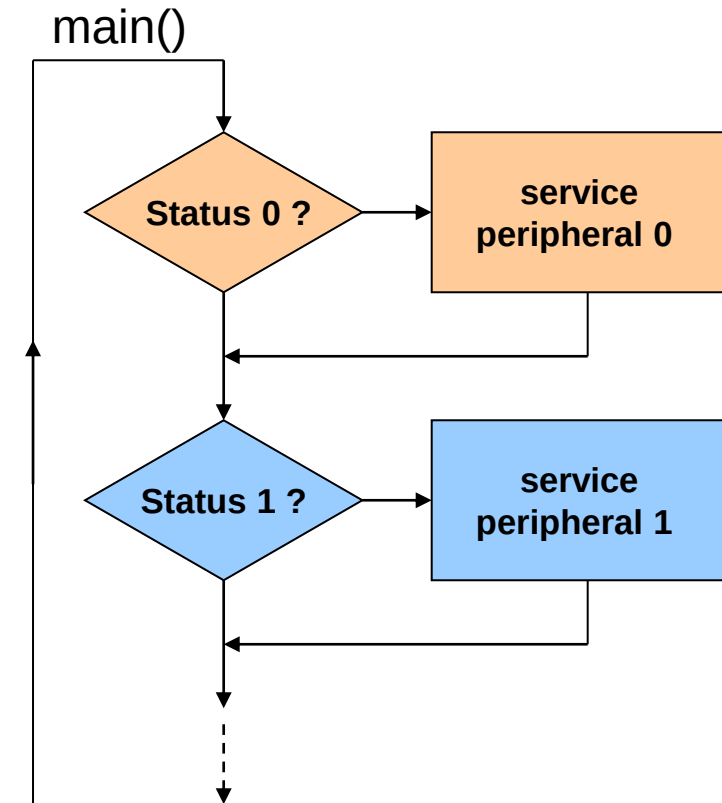


Polling

- **Periodic Query of Status Information**

- Reading of status registers in loop
- Synchronous with main program
- Advantages
 - Simple and straightforward
 - Implicit synchronization
 - Deterministic
 - No additional interrupt logic required
- Disadvantages
 - Busy wait → wastes CPU time
 - Reduced throughput
 - Long reaction times

in case of many I/O devices or
if the CPU is working on other tasks



¹⁾ to poll → abfragen

Polling

– Architecture

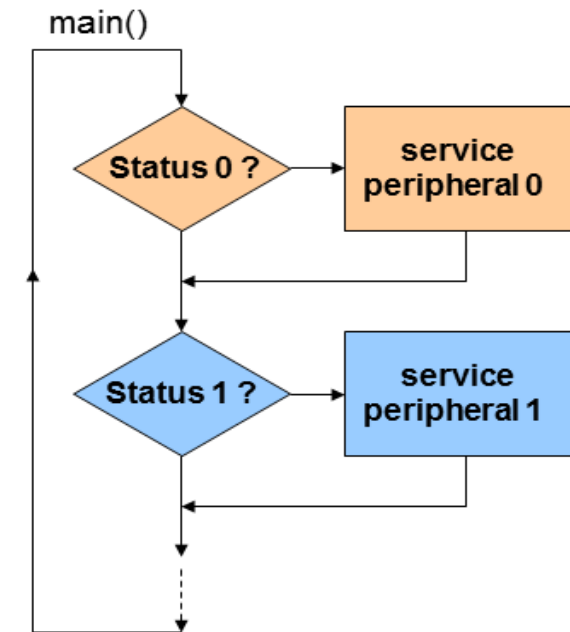
```
while(1)
{
    if ( read_byte(ADDR_BUTTON_A) ) {
        execute_task_A();
    }

    if ( read_byte(ADDR_BUTTON_B) ) {
        execute_task_B();
    }

    if ( read_byte(ADDR_BUTTON_C) ) {
        execute_task_C();
    }

    ...

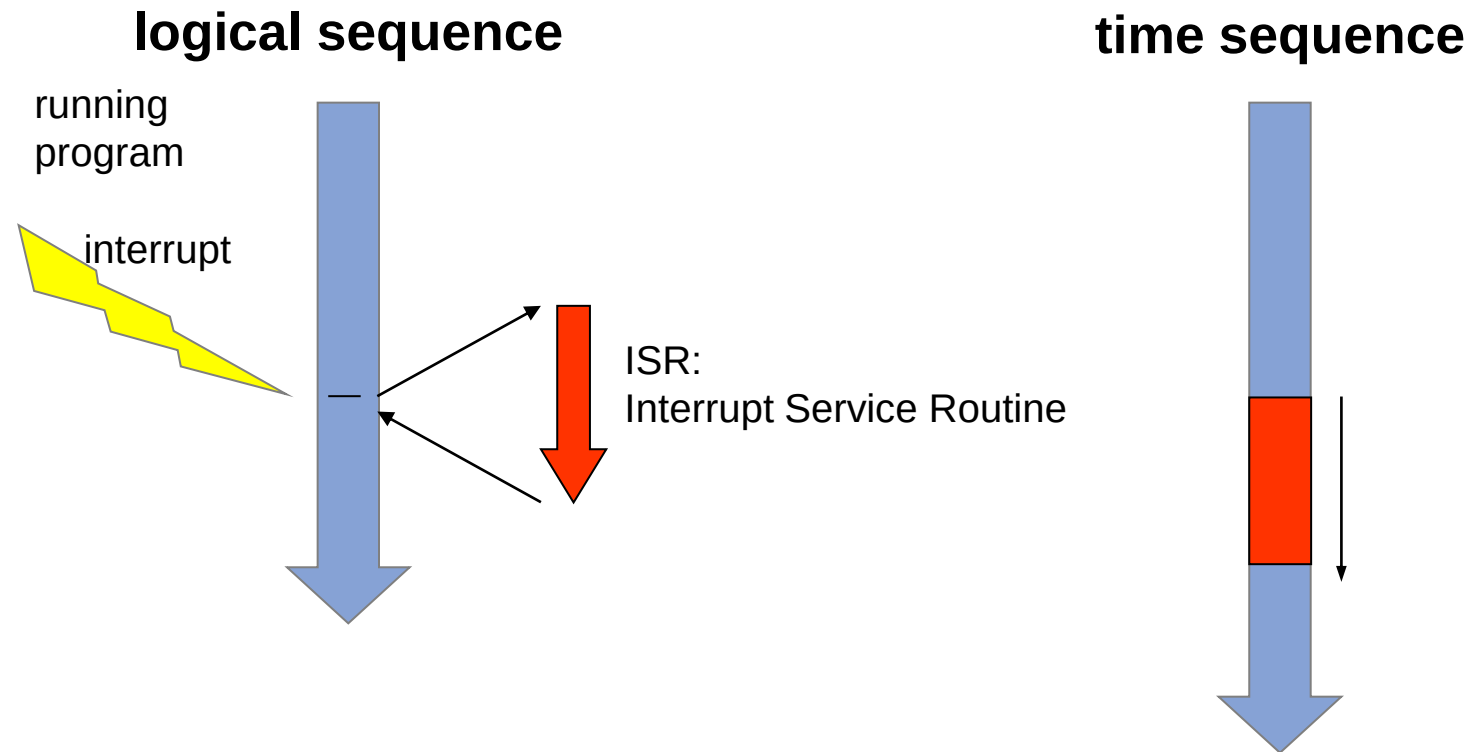
}
```



Interrupt-Driven I/O

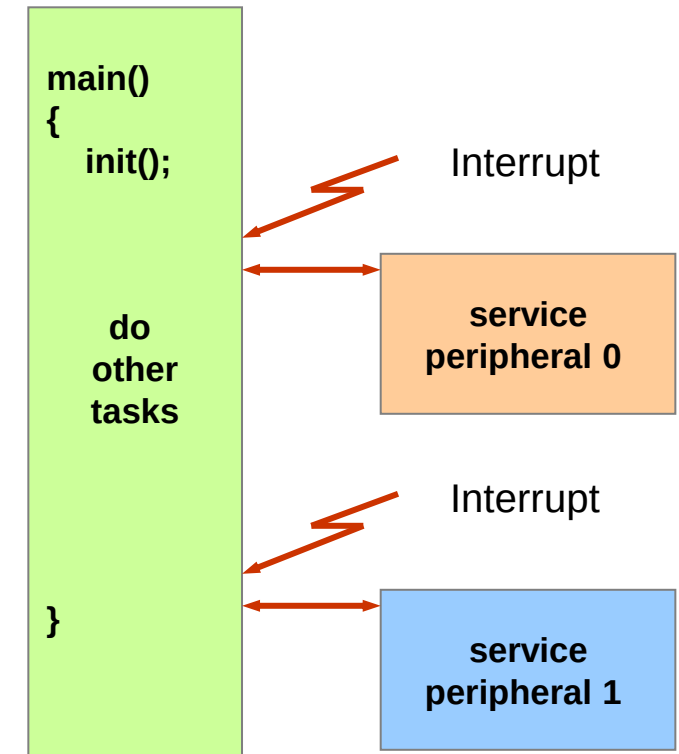
- **Alternative to polling**

Interrupt: Sudden change of program flow due to an event



Interrupt-Driven I/O

- **Main program**
 - Initializes peripherals
 - Afterwards it executes other tasks
 - Peripherals signal when they require software attention (phone call analogy)
 - Events interrupt program execution
- **Advantage**
 - No busy wait → better use of CPU time
 - Short reaction times
- **Disadvantages**
 - No synchronization (between main program and ISRs)
 - Difficult debugging



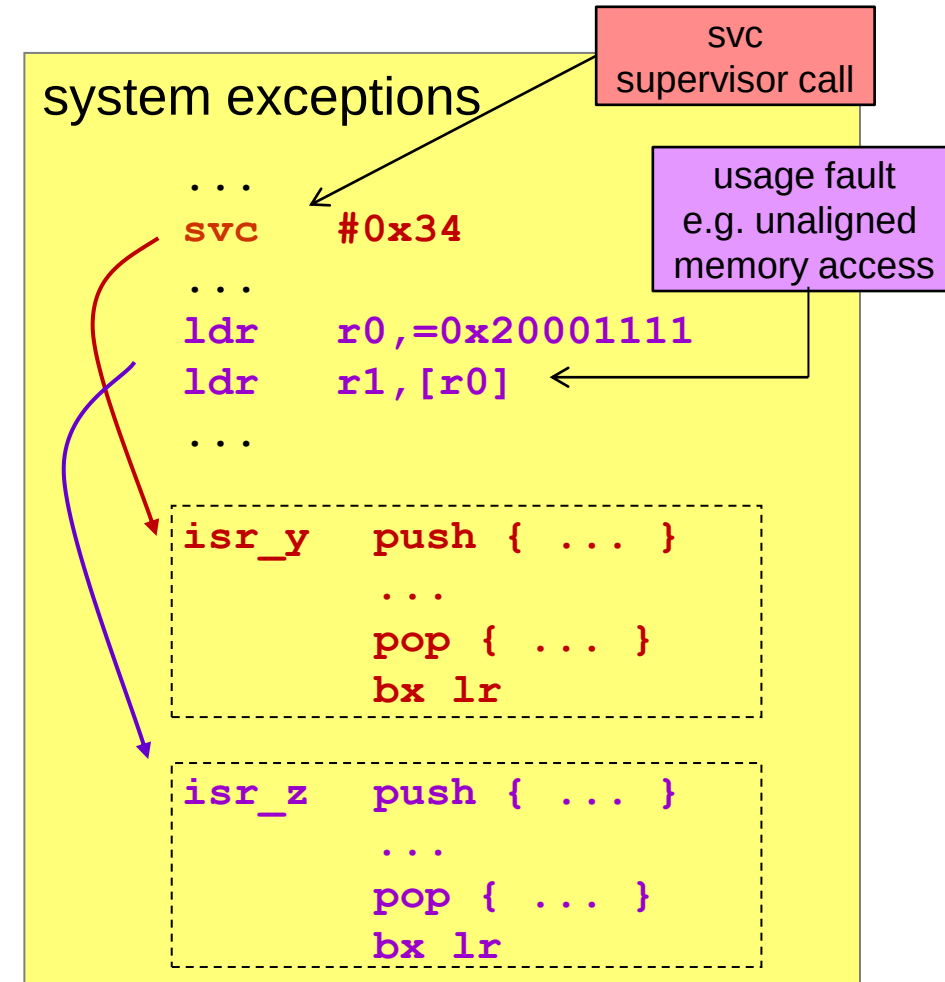
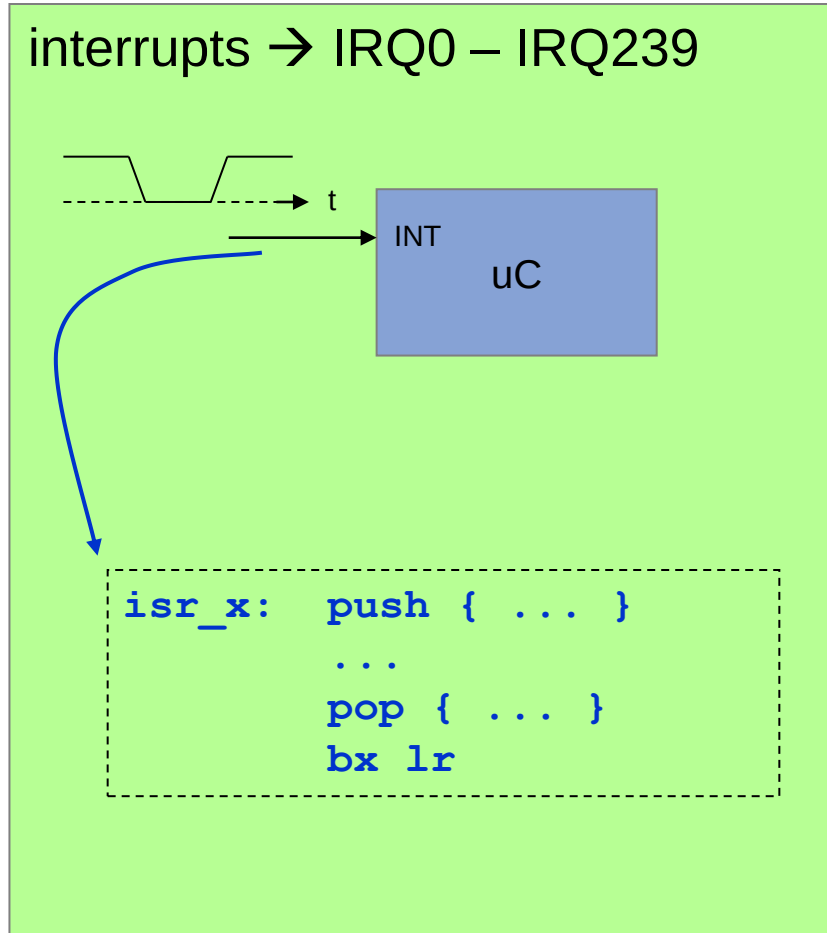
Exceptions Cortex-M3/M4

- **Interrupt sources: IRQ0 – IRQ239**
 - Peripherals¹⁾ signal to CPU that an event needs immediate attention
 - Can alternatively be generated by software request
 - Asynchronous to instruction execution
- **System exceptions**
 - Reset
 - Restart of processor
 - NMI – Non-maskable Interrupt
 - Condition that cannot be ignored, e.g. a critical hardware error
 - Faults
 - E.g. undefined instructions, unaligned accesses, etc.
 - System Level Calls
 - Operating system (OS) calls – Instruction svc and PendSV

¹⁾ GPIO (general purpose input/output), timer, serial interfaces, etc

Exceptions Cortex-M3/M4

– Examples for Exceptions



Exceptions Cortex-M3/M4

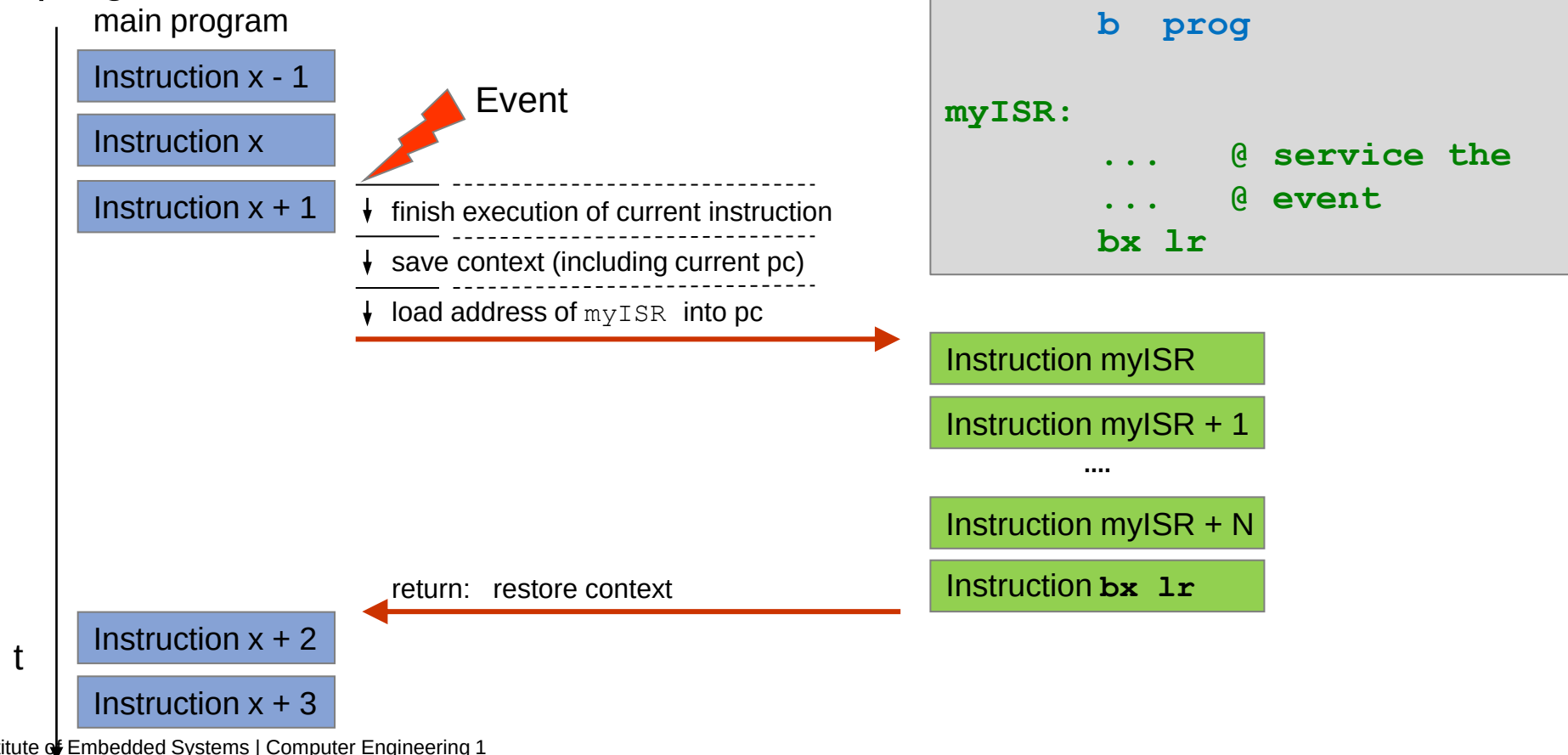
– System Exceptions

Exception Number	Exception Type	Priority	Description
1	Reset	−3 (Highest)	Reset
2	NMI	−2	Nonmaskable interrupt (external NMI input)
3	Hard Fault	−1	All fault conditions, if the corresponding fault handler is not enabled
4	MemManage Fault	Programmable	Memory management fault; MPU violation or access to illegal locations
5	Bus Fault	Programmable	Bus error; occurs when AHB interface receives an error response from a bus slave (also called <i>prefetch abort</i> if it is an instruction fetch or <i>data abort</i> if it is a data access)
6	Usage Fault	Programmable	Exceptions due to program error or trying to access coprocessor (the Cortex-M3 does not support a coprocessor)
7–10	Reserved	NA	–
11	SVCall	Programmable	System Service call
12	Debug Monitor	Programmable	Debug monitor (breakpoints, watchpoints, or external debug requests)
13	Reserved	NA	–
14	PendSV	Programmable	Pendable request for system device
15	SYSTICK	Programmable	System Tick Timer

Exceptions Cortex-M3/M4

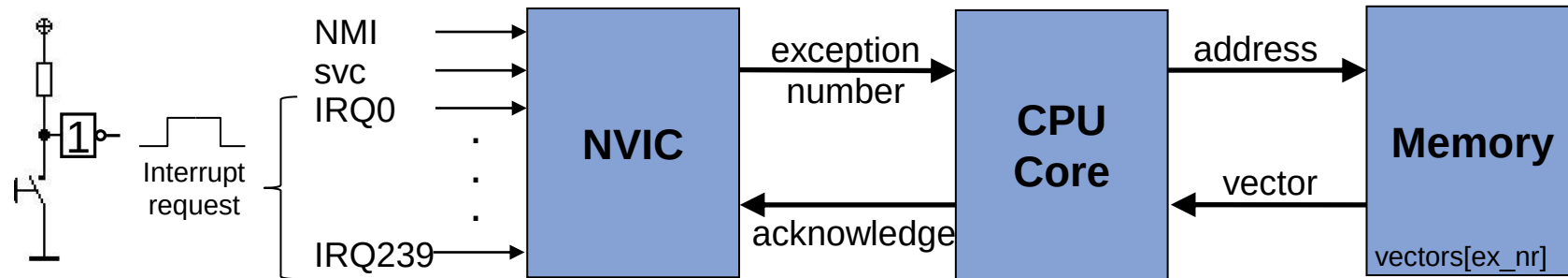
- **Interrupts: Event calls ISR**

- Change of program flow



Interrupt-System Cortex-M3/M4

- **Nested Vectored Interrupt Controller (NVIC)**
 - 240 sources can trigger exception → high level signal on IRQx
 - Forwards respective exception number to CPU



- **CPU**
 - Calculates vector table address based on exception number
 - Uses address to read vector from memory
 - Stores context on stack
 - Loads vector into pc → branch to ISR

Simultaneous exceptions from different sources will be covered later

Vector Table (Cortex-M3/M4)

- Which ISR Shall the Processor Call?
 - Each exception has a different ISR

Memory Addr.	31	0	Exception Nr.	
0x0000'0000	Top of Stack		0	System Exceptions 1 – 15
0x0000'0004	Reset		1	
0x0000'0008	NMI		2	
0x0000'000C	Hard Fault		3	
. . .	. . .		. . .	
0x0000'002C	SVC		11	
. . .	. . .		. . .	
0x0000'0038	PendSV		14	
0x0000'003C	SysTick		15	
0x0000'0040	IRQ0		16	Interrupts 0 – 239 IRQn → Exception Nr. (n + 16)
0x0000'0044	IRQ1		17	
. . .	. . .		. . .	
0x0000'03FC	IRQ239		255	

Example:
IRQ3 → Exception Nr. 19

Vector Table (Cortex-M3/M4)

– Initialization

```
; Vector Table Mapped to Address 0 at Reset
.section .Reset, "a", %progbits

0  __Vectors:  .word      __initial_sp @ Top of Stack
1              .word      Reset_Handler @ Reset Handler
2              .word      NMI_Handler @ NMI Handler
3              .word      HardFault_Handler @ Hard Fault Handler
...           .word      ...
...           .word      ...

; Interrupts
16          .word      IRQ0_Handler @ ISR for IRQ0
17          .word      IRQ1_Handler @ ISR for IRQ1
...         .word      ...
```

System Exceptions
15 vectors

Interrupts

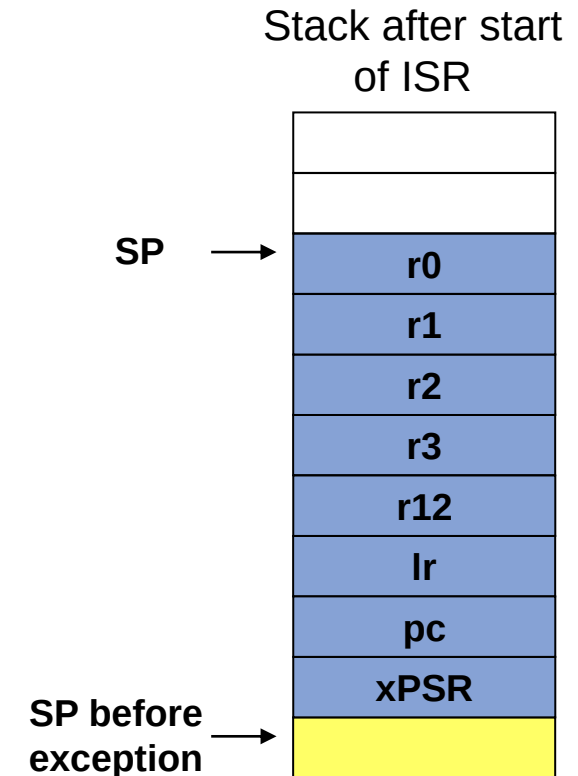
```
.section .text.myCode, "ax", %progbits
IRQ0_Handler: push { ... }

                ... @ interrupt service

                pop { ... }
                bx lr
```

Storing the Context

- **Interrupt event can take place at any time**
 - E.g. between **tst** and **beq** instructions
 - ISR call requires automatic save of flags and caller saved registers
- **ISR Call**
 - Stores xPSR, pc, lr, r12, r0 – r3 on stack
 - Stores EXC_RETURN¹⁾ to lr
- **ISR Return**
 - Use **bx lr** or **pop {..., pc}**²⁾
 - Loading EXC_RETURN¹⁾ into pc
 - restores r0 – r3, r12, lr, pc and xPSR from stack



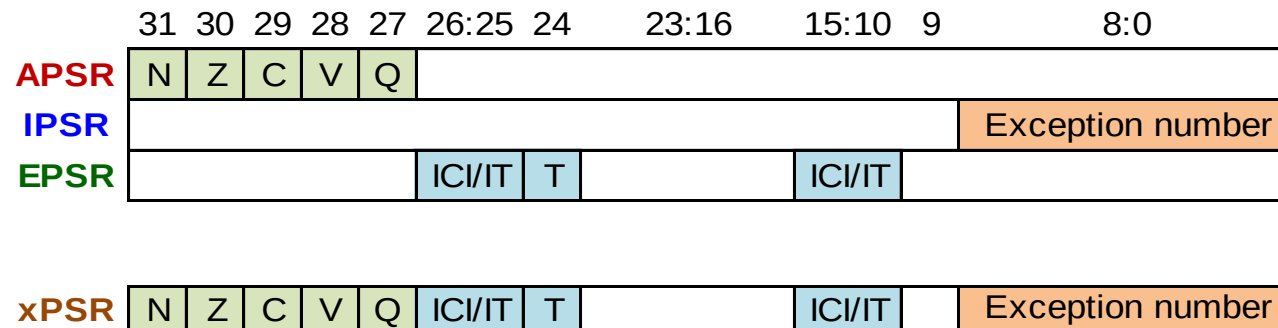
¹⁾ EXC_RETURN = 0xFFFF'FFF9

²⁾ if lr has been saved with push {..., lr} in ISR

Storing the Context

- **Program Status Registers (PSRs)**

- **APSR** Application Program Status Register
- **IPSR** Interrupt Program Status Register
- **EPSR** Execution Program Status Register

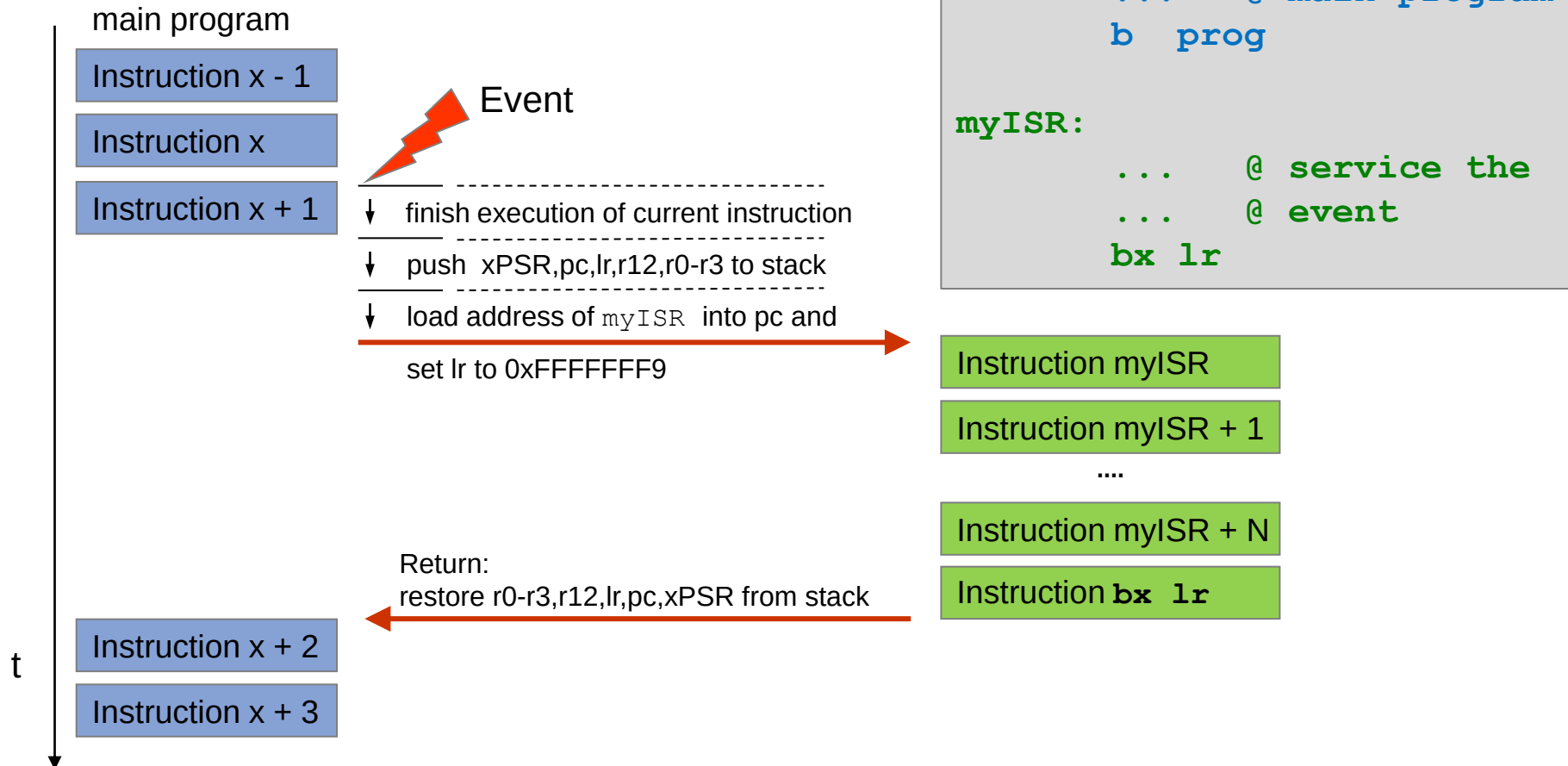


- **xPSR** Combination of all three PSRs

Exception number indicates which exception the processor is handling
ICI/IT Bits used in case of Interrupt-Continuable Instructions e.g. LDM/STM
or conditional execution of instructions (IF-THEN)
T Thumb mode. Always one

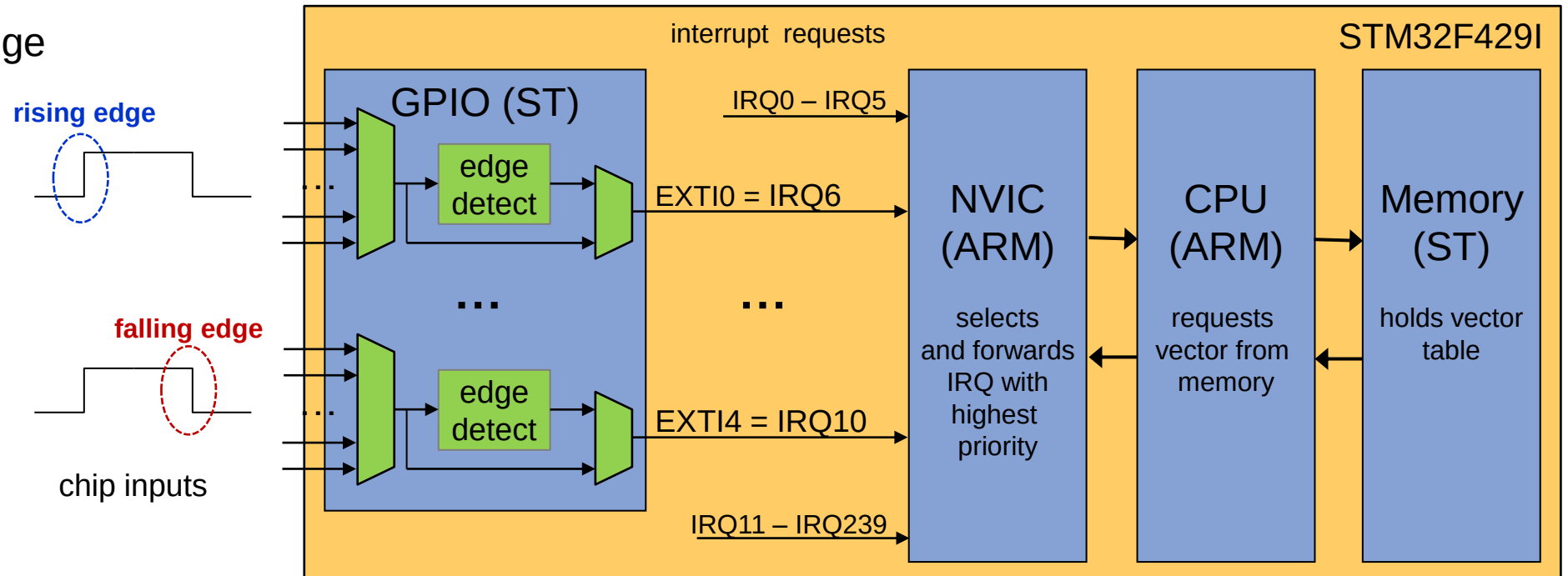
Storing the Context

– Asynchronous event calls ISR



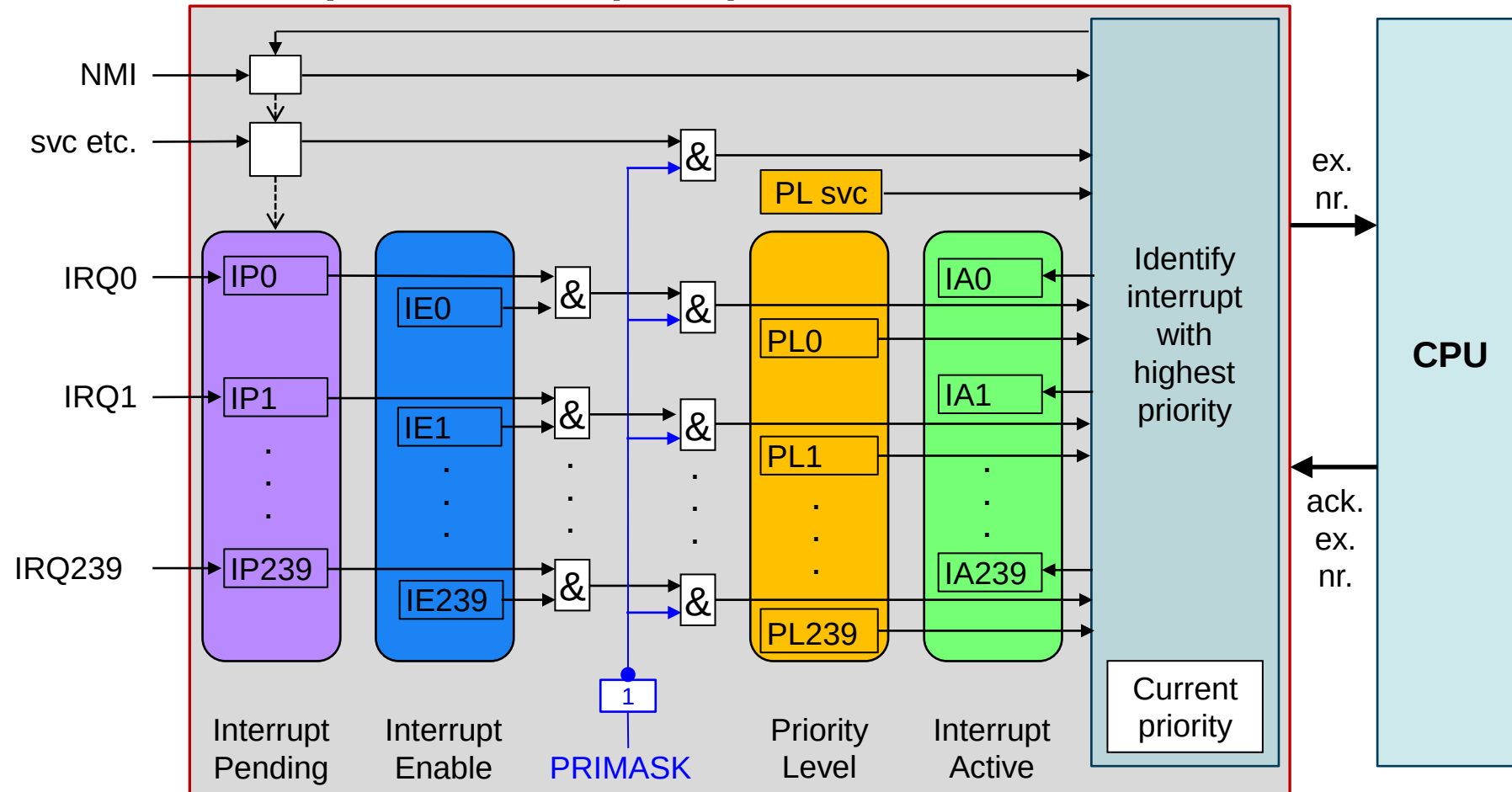
External Interrupt Pins

- **Chip Vendor (ST) adds GPIO logic**
 - EXTI0 through EXTI4 connected to IRQ6 through IRQ10
 - IRQ0 – IRQ5 / IRQ11 – IRQ239 used for other sources e.g. SPI, UART, ADC
 - Select chip input for each EXTIx line
 - Select level or edge



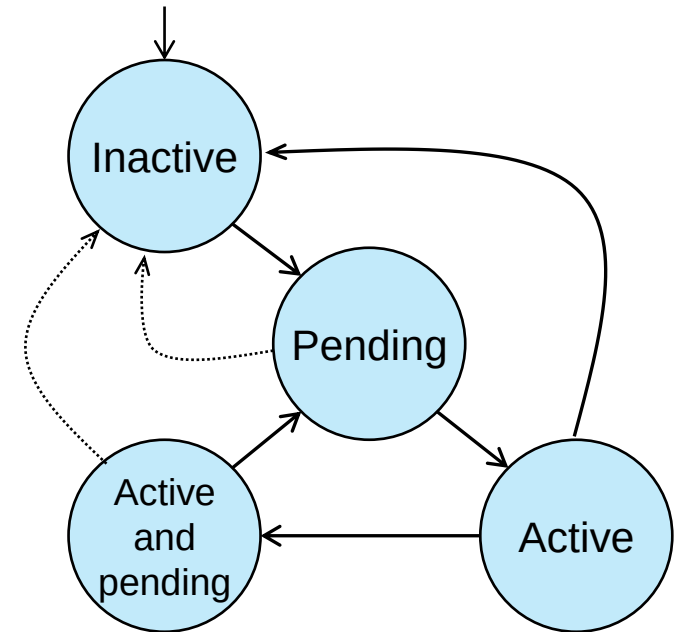
Interrupt Control

– Nested Vectored Interrupt Controller (NVIC)



Interrupt Control

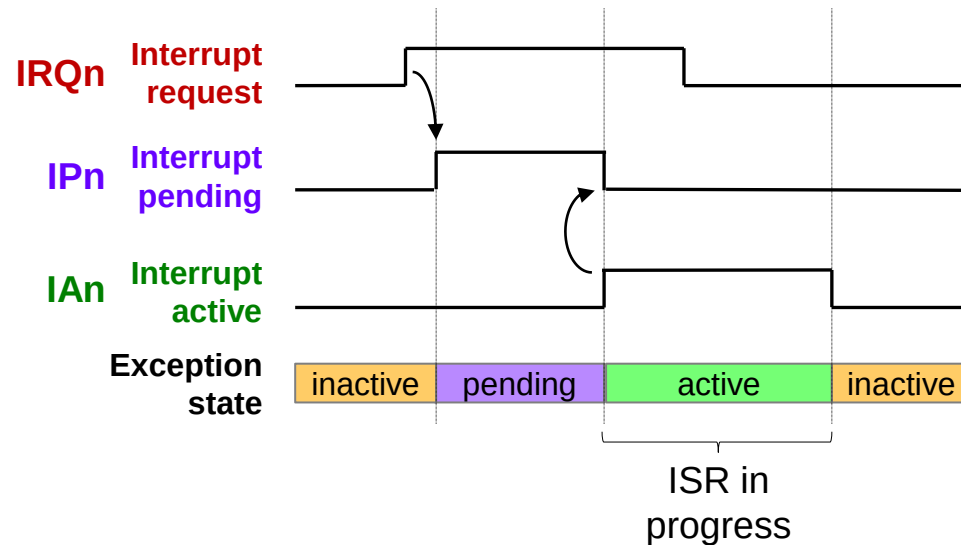
- **Exception States (for each interrupt source, e.g. IRQ17)**
 - Inactive
 - Exception is not active and not pending
 - Pending
 - Exception is waiting to be serviced by CPU
 - E.g. an interrupt event occurred ($IRQn = 1$) but interrupts are disabled (PRIMASK)
 - Active
 - Exception is being serviced by the CPU but has not completed
 - Active and pending
 - Exception is being serviced by the CPU and there is a pending exception from the same source



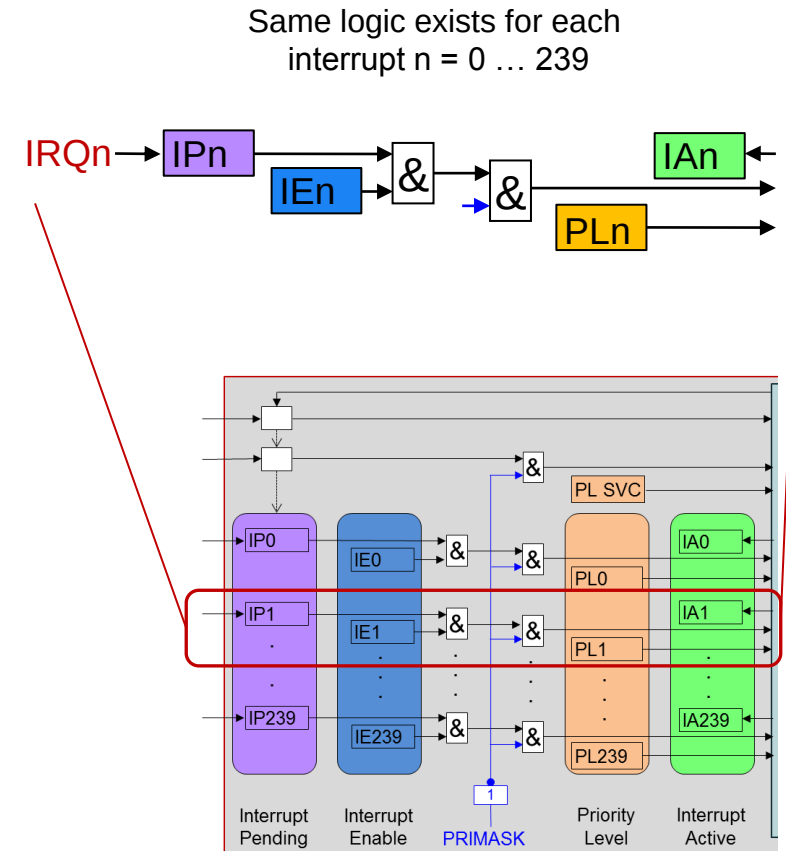
source: ARM Cortex-M3 Devices Generic User Guide

Interrupt Control

– IRQ Inputs and Pending Behavior



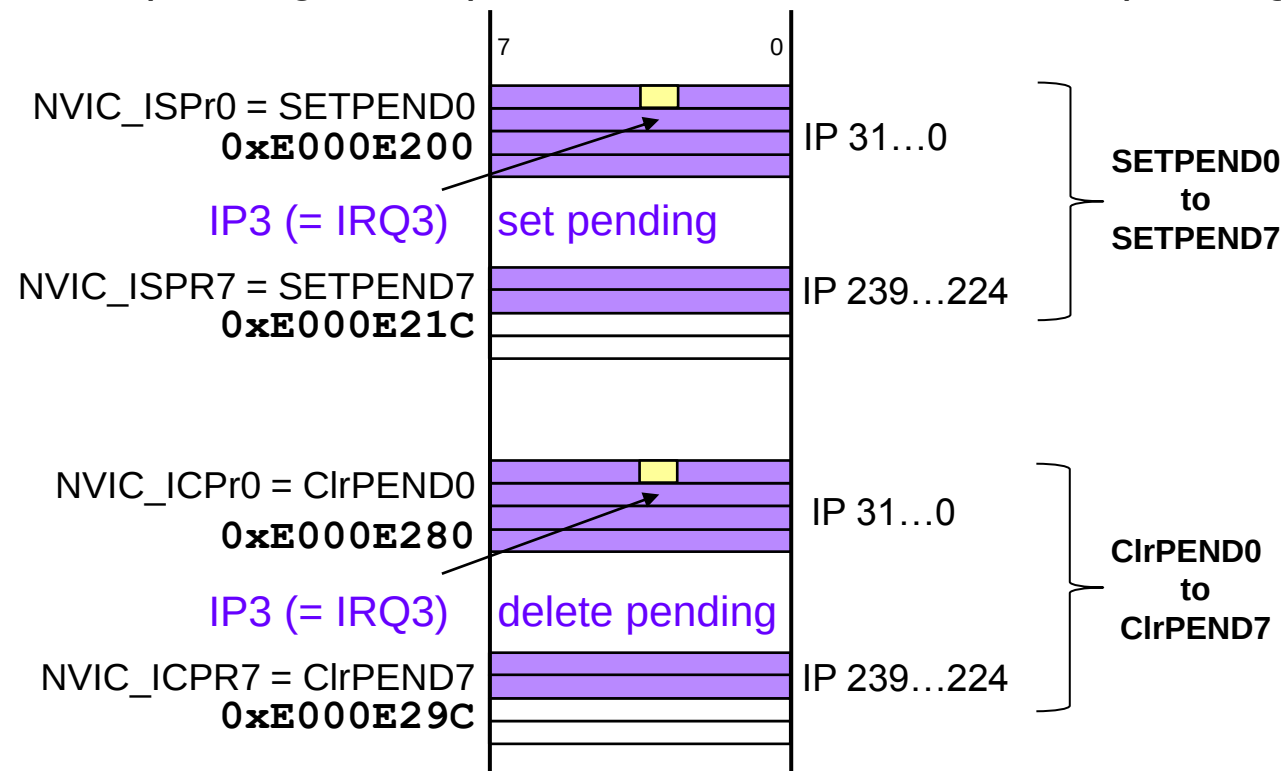
- High level on IRQn sets pending bit (IPn)
- Active Bit (IAn)
 - Set as soon as exception is being serviced
 - Resets pending bit



¹⁾ IRQn is set by hardware and usually reset by software

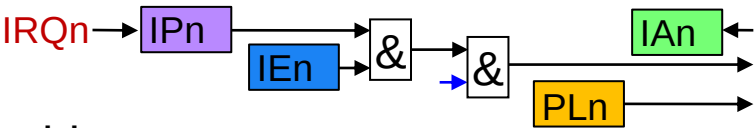
Interrupt Control

- **Interrupt Pending Registers**
 - Trigger hardware interrupt by software
 - Cancel a pending interrupt



Reading ClrPENDx by software will return the pending interrupts

- set pending bit
- clear pending bit



Trigger IRQ3 by Software

```
.equ SETPEND0, 0xE000E200
...
ldr r0,=SETPEND0
movs r1,#0x08
str r1,[r0]
```

Delete Pending IRQ3

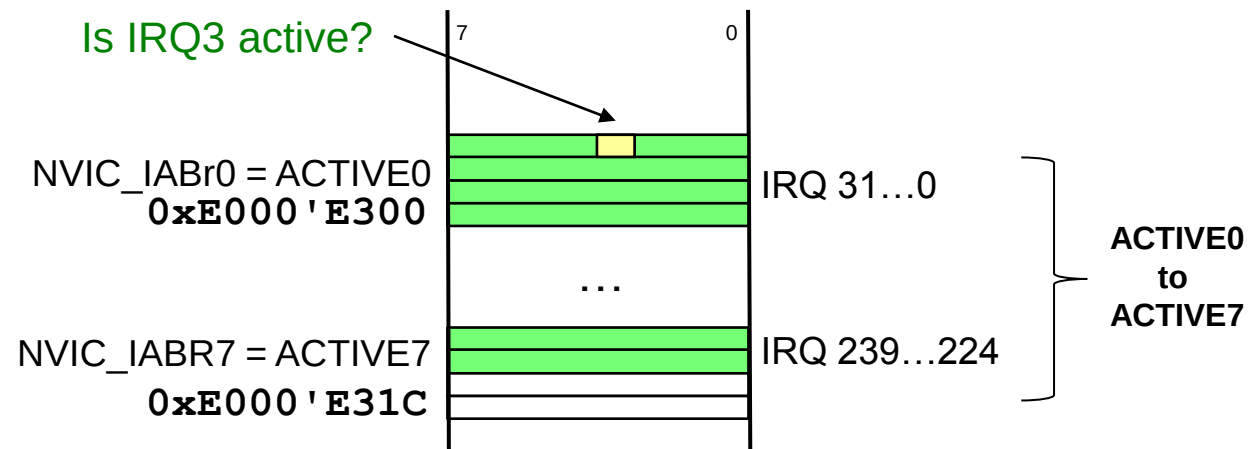
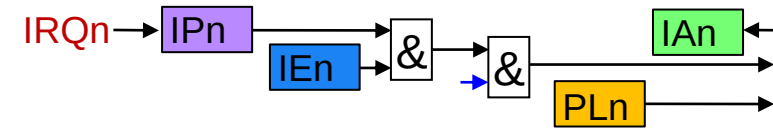
```
.equ ClrPEND0, 0xE000E280
...
ldr r0,=ClrPEND0
movs r1,#0x08
str r1,[r0]
```

ARM and ST use different names for the same register

Interrupt Control

– Interrupt Active Status Registers

- Read-only
- Corresponding bit is set when ISR starts
- Corresponding bit is cleared when interrupt return is executed



Test if IRQ3 is active

```
.equ ACTIVE0, 0xE000E300
...

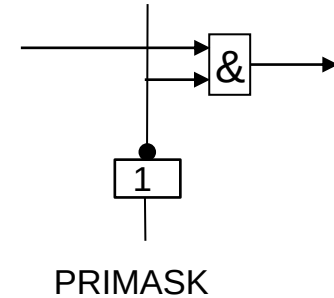
ldr    r0, =ACTIVE0
movs   r1, #0x08
ldr    r2, [r0]
tst    r1, r2
beq    ...
```

Interrupt Control

- **General Masking of Interrupts**

- PRIMASK

- Single bit controlling all maskable interrupts



- Disable set PRIMASK
 - Enable clear PRIMASK

Assembly

```
cpsid1) i  
cpsie1) i
```

C

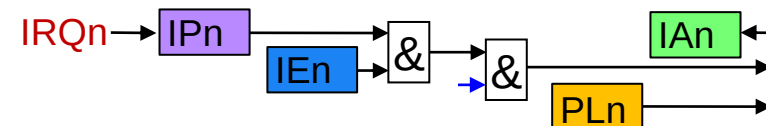
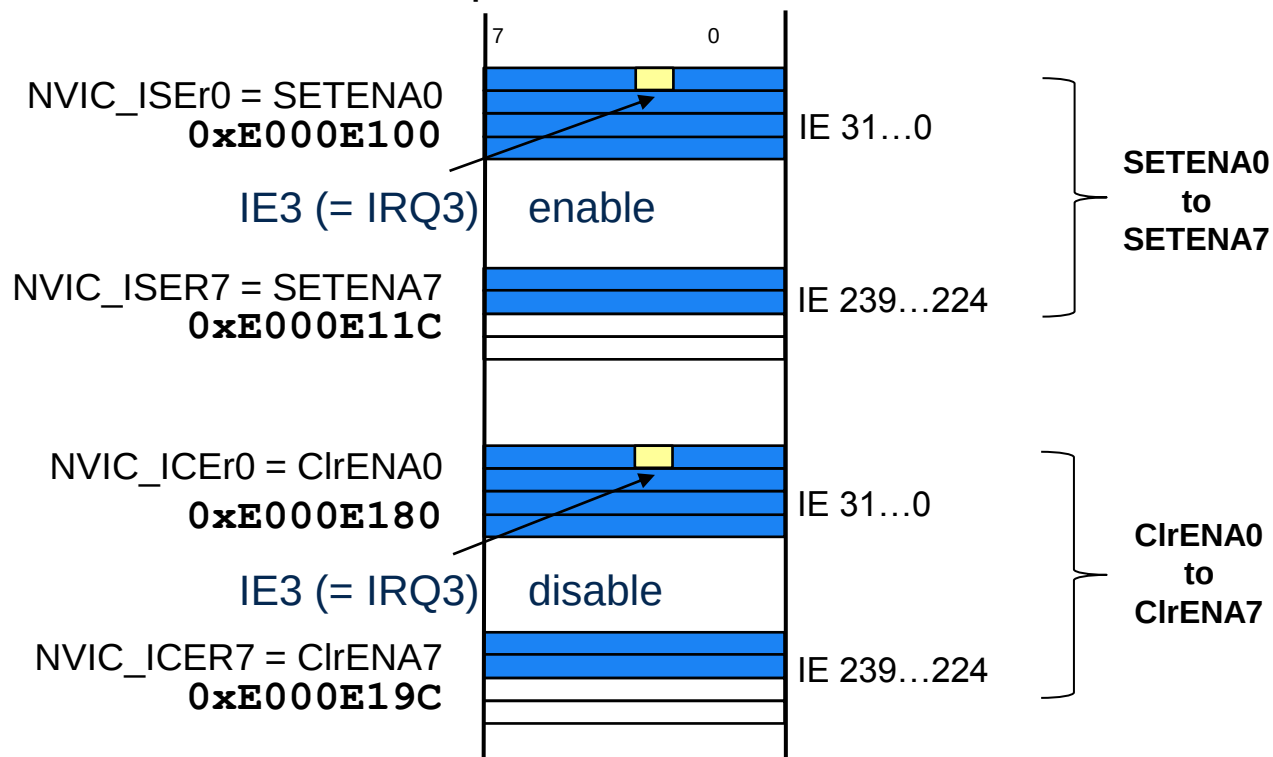
```
__disable_irq();  
__enable_irq();
```

- On reset PRIMASK = 0 → enabled
 - **Non-maskable Interrupt (NMI)**
 - Power-fail, emergency button, watchdog, ...

¹ cpsie/D = Change Processor State Interrupt Enable / Disable

Interrupt Control

- **Interrupt Enable Registers**
 - Individual masking of interrupt sources
 - IEn cleared pending bit not forwarded
 - IEn set interrupt enabled



ClrENA and SETENA registers can be read by software and will return the settings of the IE bits

Enable IRQ3

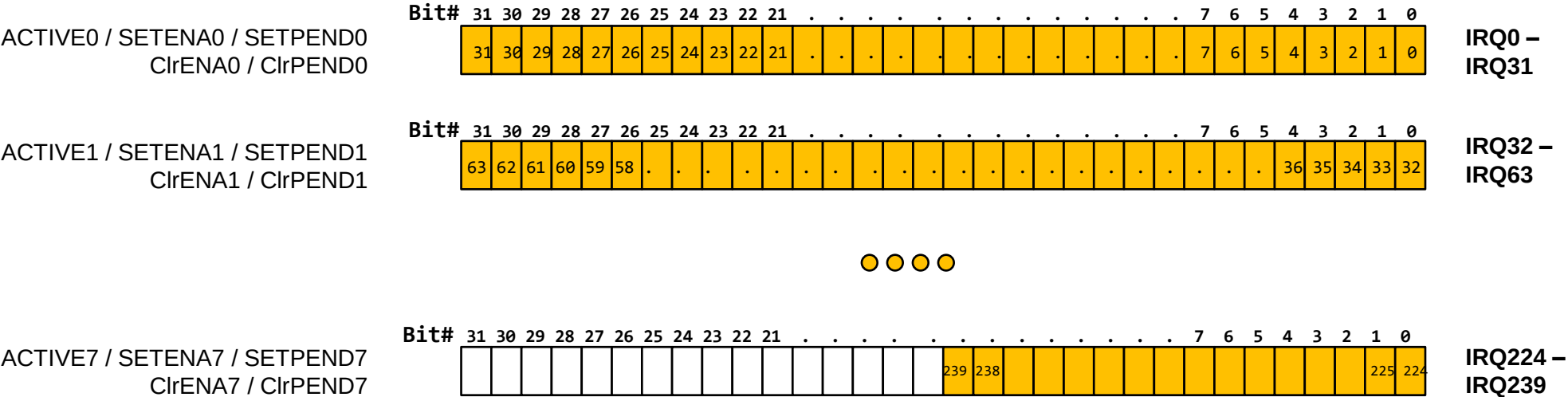
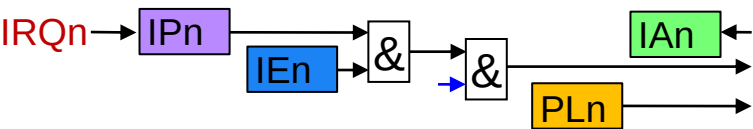
```
.equ SETENA, 0xE000E100
...
ldr    r0, =SETENA0
movs   r1, #0x08
str    r1, [r0]
```

Disable IRQ3

```
.equ ClrENA0, 0xE000E180
...
ldr    r0, =ClrENA0
movs   r1, #0x08
str    r1, [r0]
```

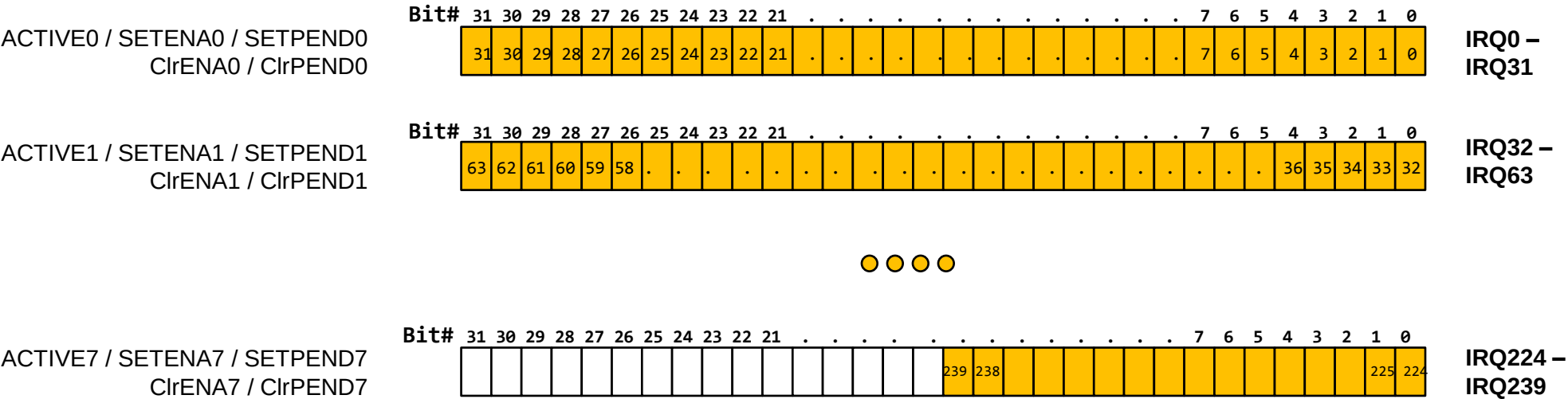
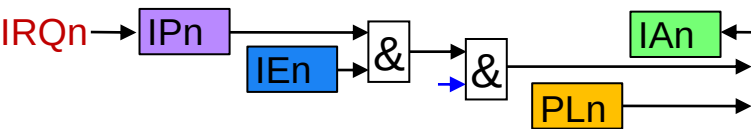
Interrupt Control – Register View

- 1 Bit for each IRQ
 - Registers are written as 32-bit words
 - Only bits with value “1” have an effect, “0” is ignored
 - Separate registers to read and write



Interrupt Control – Register View

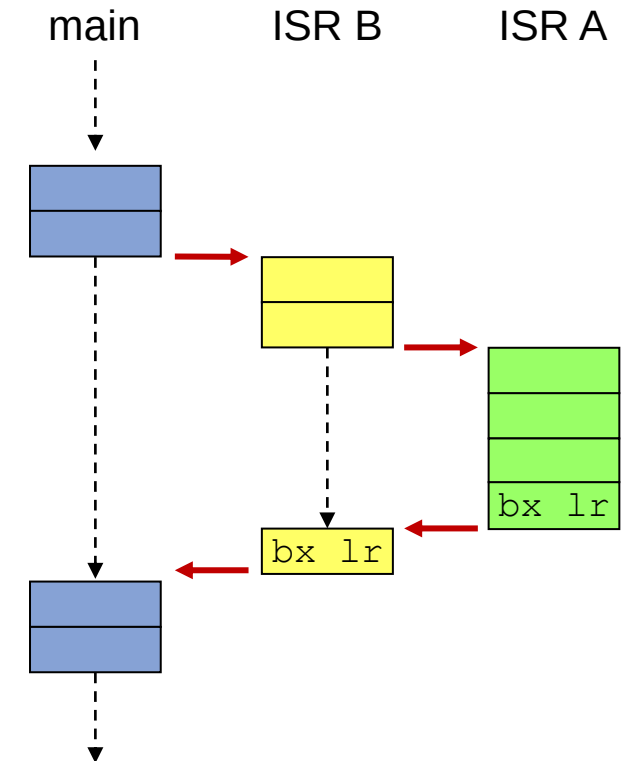
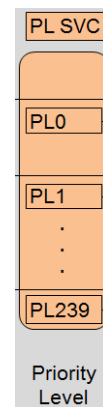
- **Exercise:**
 - Enable IRQ 46 and clear pending IRQ 129?



Nested Exceptions

– Preemption

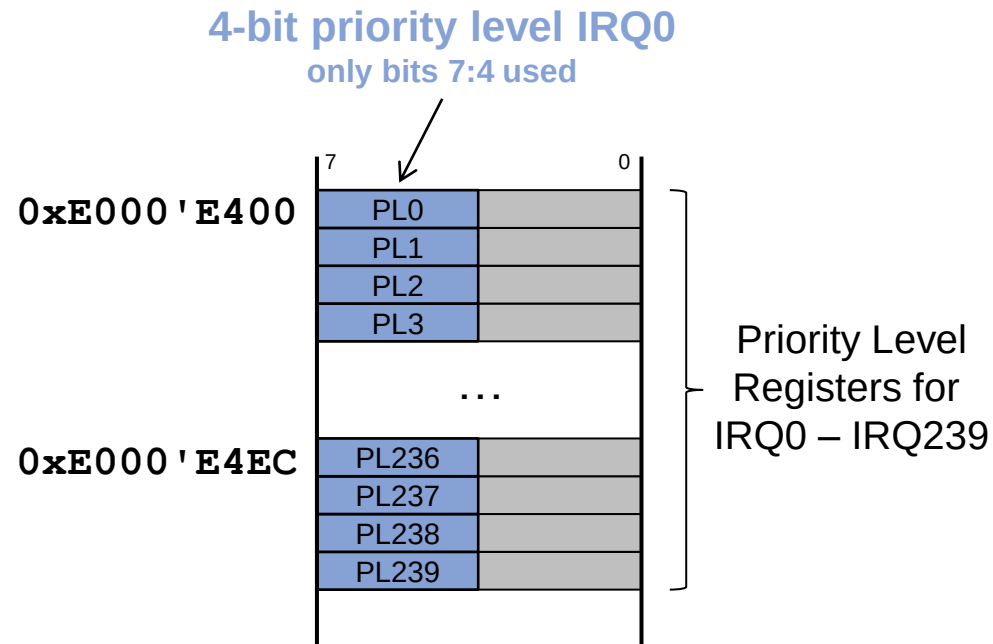
- Service routine A temporarily interrupts service routine B
- Assigned priority level for each exception
 - Levels define whether A can preempt B
- Fixed priorities
 - Reset (-3), NMI (-2), hard fault (-1)
- All other priorities are programmable



if A has been programmed to higher priority than B

Nested Exceptions

- **Priority Level Registers**
 - The lower a priority level, the greater the priority
 - 4-bit priority level ¹⁾ 0x0 – 0xF



Set priority of IRQ3 to 5

```
.equ PL_IRQ3, 0xE000E403
...

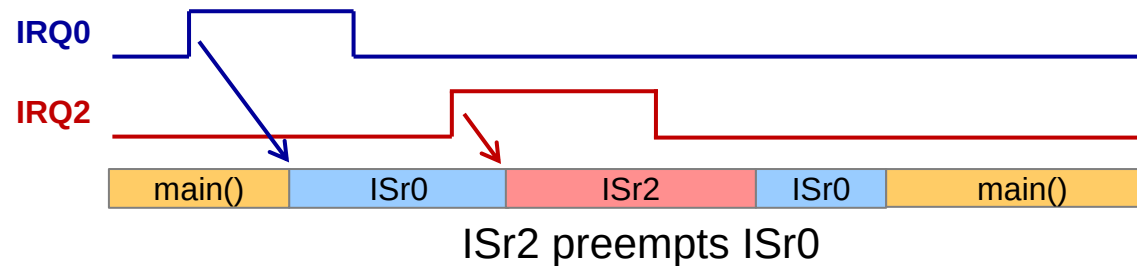
ldr    r0, =PL_REG_IRQ3
movs   r1, #0x50
strb   r1, [r0]
```

priority level registers for system exceptions
0xE000'ED18 – 0xE000'ED23

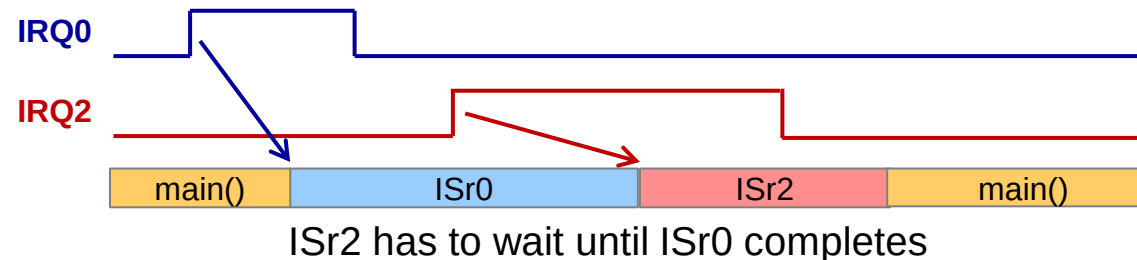
1) Cortex-M3/4 can handle up to 8 bits but STM uses only 4
Registers are called NVIC_IPRx on ST

Nested Exceptions

- **IRQ request while other ISR is running**
 - ISr0 is executing (triggered by IRQ0)
 - Interrupt request IRQ2 arrives
 - $PL2 < PL0 \rightarrow$ IRQ2 has higher priority than IRQ0



- $PL2 \geq PL0 \rightarrow$ IRQ2 has lower or same priority than IRQ0



Nested Exceptions

- **Two or more pending interrupts**
 - (1) IRQ requests arrive simultaneously
 - (2) More than one request has been waiting for a higher priority ISR to complete
 - NVIC will evaluate the priorities of the pending interrupts
 - Selects IRQx with highest priority i.e.
 - with lowest priority level value (PL)
 - with lowest number (in case of identical PL values)
 - Other interrupts remain pending

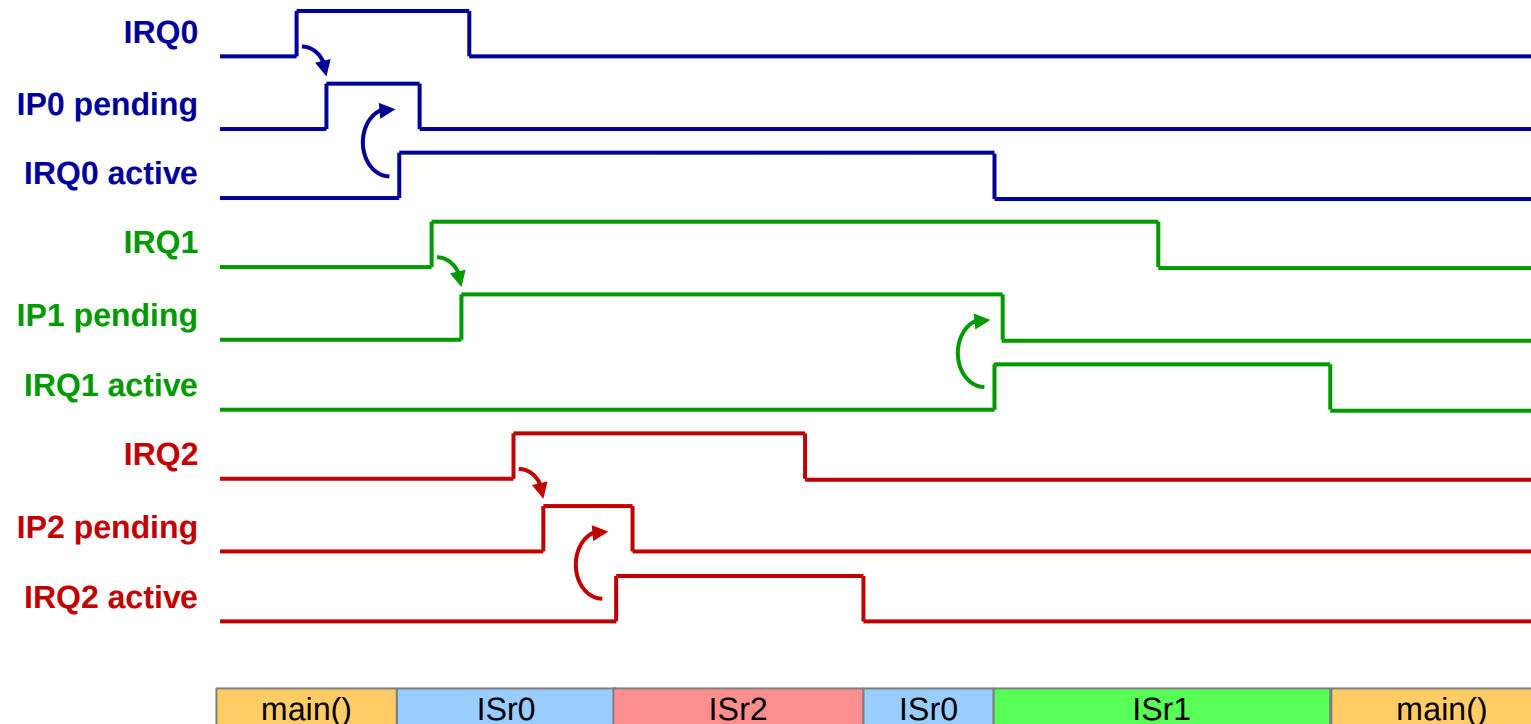
Nested Exceptions

– Example Priorities

- ISr1 does not preempt ISr0
- ISr2 preempts ISr0

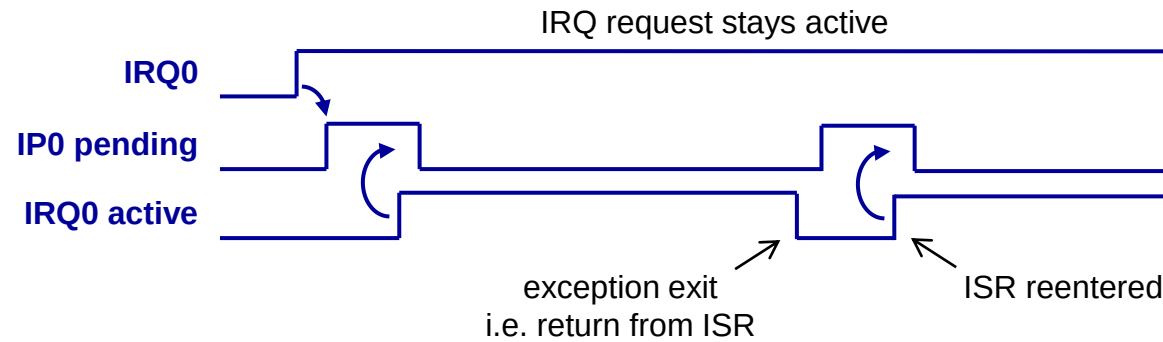
assuming

IRQ0	PL0 = 0x2	medium priority
IRQ1	PL1 = 0x3	lowest priority
IRQ2	PL2 = 0x1	highest priority



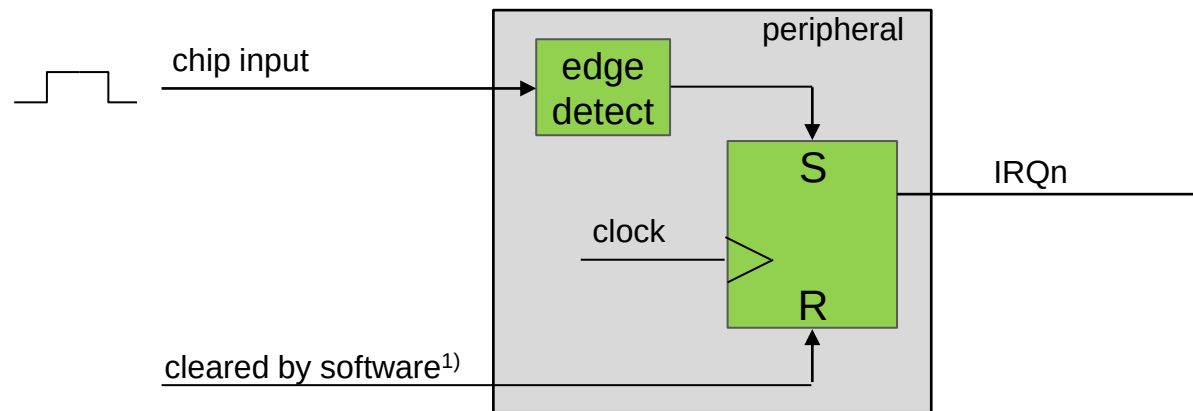
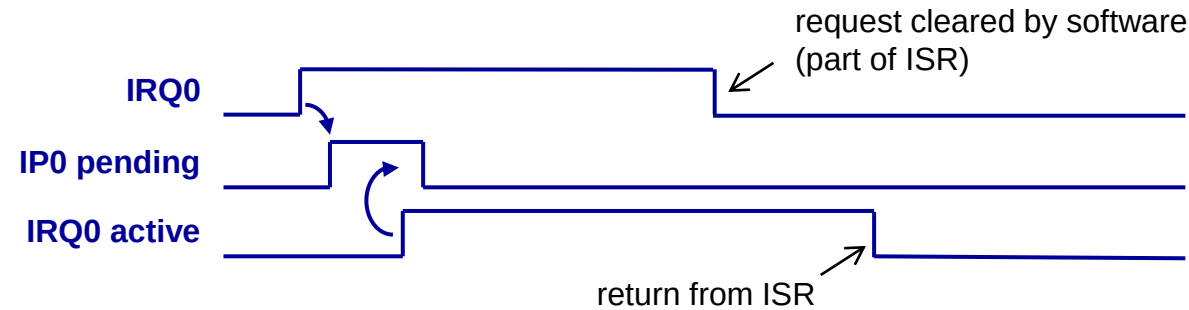
Special Interrupt Situations

- **IRQ request stays active**
 - Interrupt becomes pending again



Special Interrupt Situations

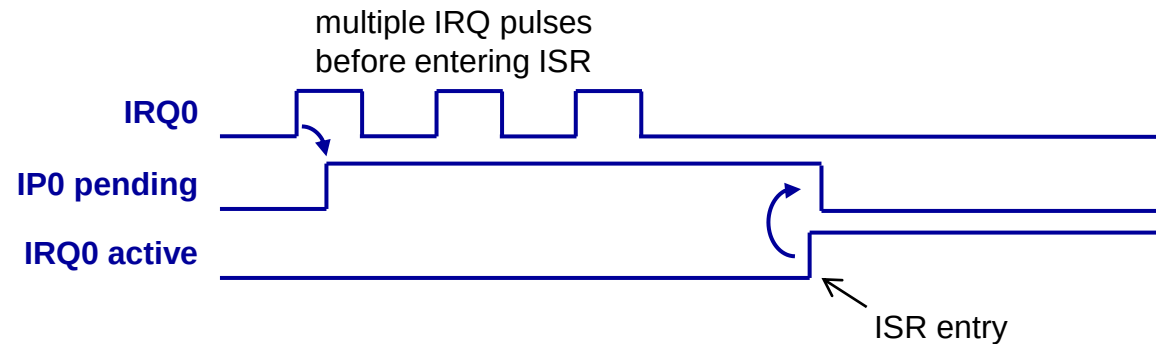
- **Common Configuration on Microcontrollers**
 - IRQ Set by Hardware – Cleared by Software



1) The software clears the bit at the correct position at the defined register address

Special Interrupt Situations

- **Multiple IRQ request pulses before entering ISR**
 - Treated as single interrupt
 - Events are lost



CMSIS

- **What is CMSIS?**
 - Cortex Microcontroller Software Interface Standard
 - Vendor-independent hardware abstraction layer for Cortex-M
 - Defines generic tool interfaces
 - ISO/IEC C code cannot directly access some Cortex-M3 instructions
 - intrinsic functions required

CMSIS Functions

– NVIC Control

```
void NVIC_EnableIRQ (IRQn_t IRQn)
```

Enable IRQn

```
void NVIC_DisableIRQ (IRQn_t IRQn)
```

Disable IRQn

```
uint32_t NVIC_GetPendingIRQ (IRQn_t IRQn)
```

Return true (IRQ-Number) if IRQn is pending

```
void NVIC_SetPendingIRQ (IRQn_t IRQn)
```

Set IRQn pending

```
void NVIC_ClearPendingIRQ (IRQn_t IRQn)
```

Clear IRQn pending status

```
uint32_t NVIC_GetActive (IRQn_t IRQn)
```

Return '1' if active bit of IRQn is set. '0' otherwise

```
void NVIC_SetPriority (IRQn_t IRQn, uint32_t priority)
```

Set priority for IRQn

```
uint32_t NVIC_GetPriority (IRQn_t IRQn)
```

Read priority of IRQn

```
void NVIC_SystemReset (void)
```

Reset the system

Data Consistency

– Example

```
typedef struct {
    uint8_t minutes;
    uint8_t seconds;
} time_t;

static time_t time = { 0, 0 };

int main(void)
{
    while (1) {

        write_byte(ADDR_LED_7_0,
                   time.seconds);
        write_byte(ADDR_LED_15_8,
                   time.minutes);

    }
}
```

```
void IRQ1_Handler(void)
{
    time.seconds++;
    if (time.seconds > 59) {
        time.seconds = 0;
        time.minutes++;
    }
}
```

Data Consistency

– Example

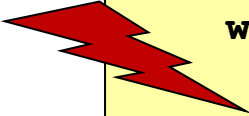
```
typedef struct {
    uint8_t minutes;
    uint8_t seconds;
} time_t;

static time_t time = { 0, 0 };

int main(void)
{
    while (1) {

        write_byte(ADDR_LED_7_0,
                   time.seconds);
        write_byte(ADDR_LED_15_8,
                   time.minutes);

    }
}
```



```
void IRQ1_Handler(void)
{
    time.seconds++;
    if (time.seconds > 59) {
        time.seconds = 0;
        time.minutes++;
    }
}
```

Data Consistency

– Example

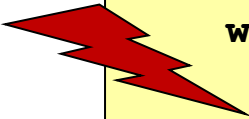
```
typedef struct {
    uint8_t minutes;
    uint8_t seconds;
} time_t;

static time_t time = { 0, 0 };

int main(void)
{
    while (1) {

        write_byte(ADDR_LED_7_0,
                    time.seconds);
        write_byte(ADDR_LED_15_8,
                    time.minutes);

    }
}
```



```
void IRQ1_Handler(void)
{
    time.seconds++;
    if (time.seconds > 59) {
        time.seconds = 0;
        time.minutes++;
    }
}
```

time = { 15, 59 }

- 1) Output 59 → LED_7_0
- 2) Interrupt → time = { 16, 0 }
- 3) Output 16 → LED_15_8

→ display: 16 59 !!!

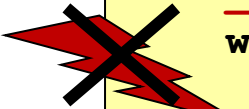
Data Consistency

– Example

```
typedef struct {
    uint8_t minutes;
    uint8_t seconds;
} time_t;

static time_t time = { 0, 0 };

int main(void)
{
    while (1) {
        __disable_irq();
        write_byte(ADDR_LED_7_0,
                    time.seconds);
        write_byte(ADDR_LED_15_8,
                    time.minutes);
        __enable_irq();
    }
}
```



```
void IRQ1_Handler(void)
{
    time.seconds++;
    if (time.seconds > 59) {
        time.seconds = 0;
        time.minutes++;
    }
}
```

time = { 15, 59 }
 1) Output 59 → LED_7_0
 2) Interrupt → time = { 16, 0 }
 3) Output 16 → LED_15_8
 → display: 16 59 !!!

Data Consistency

- Data structure must not be changed during output
- Disable Interrupts during output
- **Multitasking problem**

Conclusions

- **Polling vs. Interrupt-driven I/O**
- **Exceptions**
 - System exceptions: Reset, NMI, Faults, System Level Calls
 - Interrupts IRQ0 – IRQ239
- **Nested Vectored Interrupt Controller**
 - Vector Table holds addresses of interrupt service routines (ISR)
 - Save context including xPSR
 - Masking of interrupts
 - Pending and active bits
 - Priority levels
 - IRQn: Often hardware set; software reset
- **Data Consistency Issues**